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Dr. Kim's Fall 2026 MTH 207: HW06 (portion to submit). **Please look online at the FULL version.**
Due Thursday October 22 (official). Due Thursday October 29 (actual)

1. If $y = \sin(x^8 + 8^x)$ find y'

$y' = \cos(x^8 + 8^x) \cdot (8x^7 + 8^x \ln 8)$

2. If $y = \sin(\sin(\sin(\arctan(x))))$ find $\frac{dy}{dx}$

$\frac{dy}{dx} = \cos(\sin\sin(\arctan(x))) \cdot \cos(\sin(\arctan(x))) \cdot \cos(\arctan x) \cdot \frac{1}{1+x^2}$

3. Find $\frac{dy}{dx}$ if y depends on x according to $xy + x = y + \sin(x^3y^8)$

$$x \cdot \frac{dy}{dx} + y \cdot 1 + 1 = \frac{dy}{dx} + \cos(x^3y^8) \cdot \left(x^3 \cdot 8y^7 \frac{dy}{dx} + y^8 \cdot 3x^2\right)$$

$$x \cdot \frac{dy}{dx} + y \cdot 1 + 1 = \frac{dy}{dx} + \cos(x^3y^8) \cdot x^3 \cdot 8y^7 \frac{dy}{dx} + \cos(x^3y^8) \cdot y^8 \cdot 3x^2$$

$$x \cdot \frac{dy}{dx} - \frac{dy}{dx} - \cos(x^3y^8) \cdot x^3 \cdot 8y^7 \frac{dy}{dx} = \cos(x^3y^8) \cdot y^8 \cdot 3x^2 - y - 1$$

$$\frac{dy}{dx}\left(x - 1 - \cos(x^3y^8) \cdot x^3 \cdot 8y^7\right) = \cos(x^3y^8) \cdot y^8 \cdot 3x^2 - y - 1$$

$$\frac{dy}{dx} = \frac{\cos(x^3y^8) \cdot y^8 \cdot 3x^2 - y - 1}{x - 1 - \cos(x^3y^8) \cdot x^3 \cdot 8y^7}$$

4. Find an equation to the graph of $y = \arctan(x)$ at $x = 1$

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(1) = \frac{1}{1+1^2} = \frac{1}{2}$$

$$f(1) = \arctan(1) = \frac{\pi}{4}$$

$$y - \frac{\pi}{4} = \frac{1}{2}(x - 1)$$